

EXPERIMENT: 9

MASSIVE MIMO CAPACITY VS SNR AND GAIN USING MATLAB

Objective – 1 Plot of Channel Capacity (bps/Hz) versus SNR (dB).

MATLAB Code:

```
clc;

clear;

close all;

snr = 0:5:30;

snr_linear = 10.^(snr/10);

capacity = log2(1 + snr_linear);

figure;

plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);

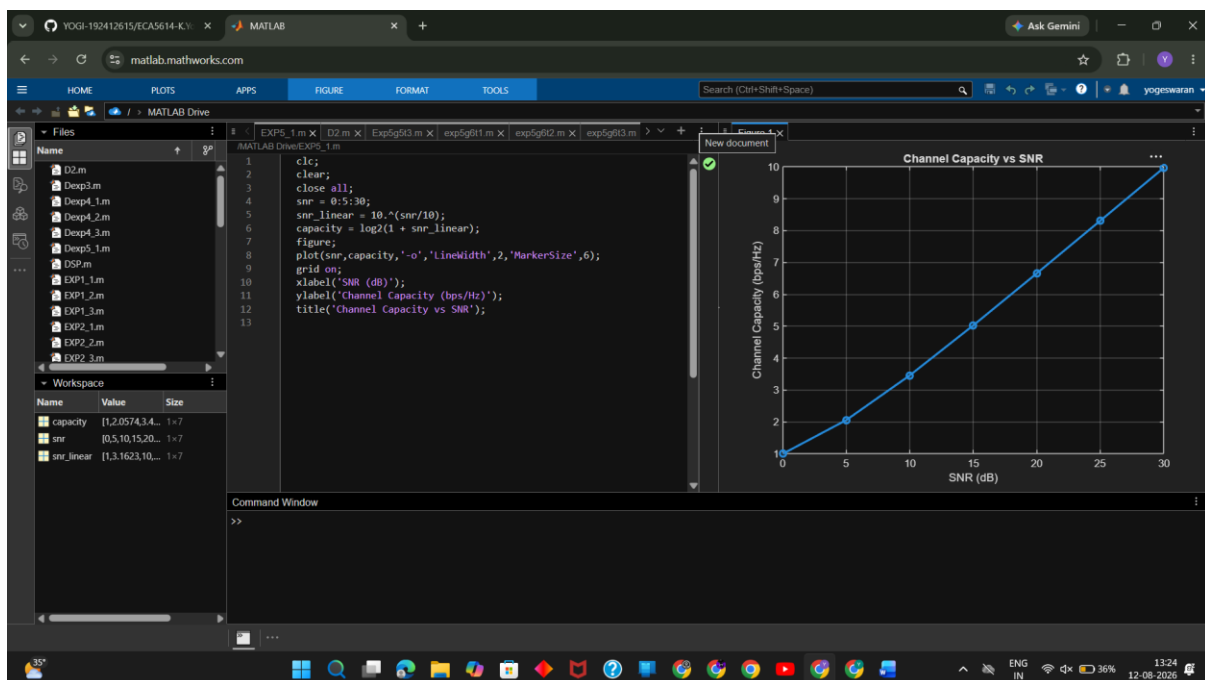
grid on;

xlabel('SNR (dB)');

ylabel('Channel Capacity (bps/Hz)');

title('Channel Capacity vs SNR');
```

OUTPUT



Objective – 2 Bar graph comparing SNR values under different communication scenarios.

Matlab Code :

```
clc;

clear;

close all;

snr = 0:5:30;

snr_linear = 10.^(snr/10);

capacity = log2(1 + snr_linear);

figure;

plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);

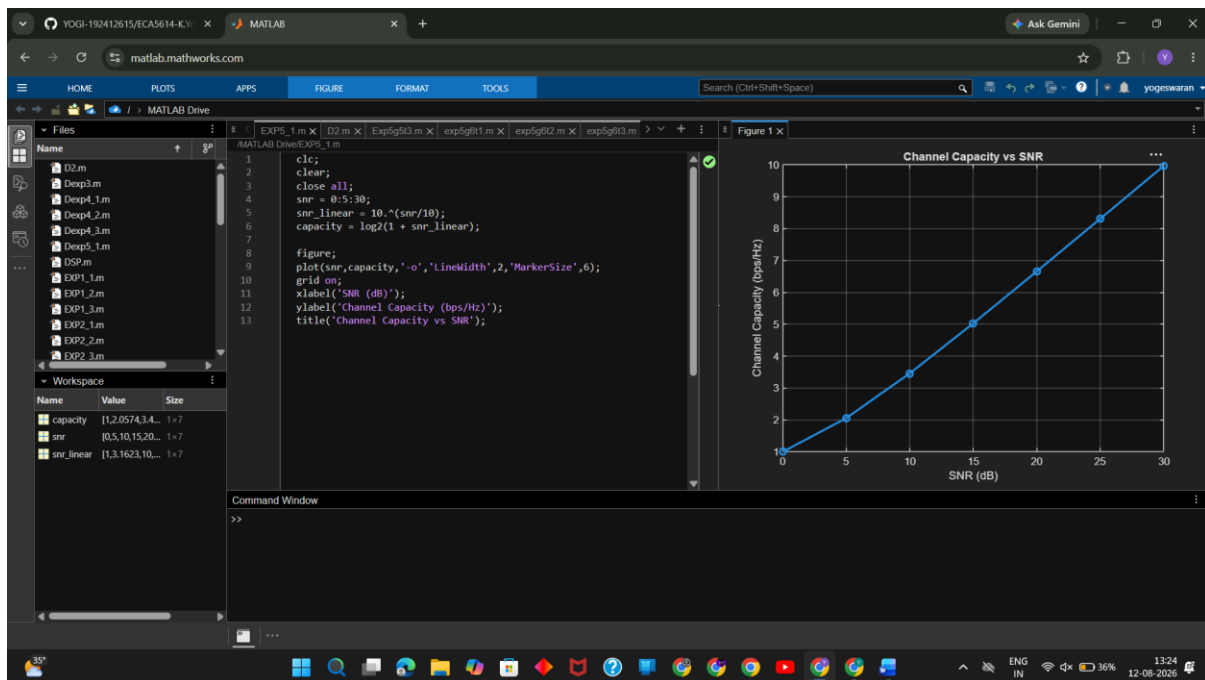
grid on;

xlabel('SNR (dB)');

ylabel('Channel Capacity (bps/Hz)');

title('Channel Capacity vs SNR');
```

OUTPUT



Objective – 3 Bar graph showing Antenna Gain for different antenna array sizes.

MATLAB Code:

```
clc;

clear;
```

```
close all;
```

```
antennas = [4 8 16 32 64]; % Number of Antenna Elements
```

```
gain = 10*log10(antennas); % Antenna Gain (dB)
```

```
figure
```

```
bar(antennas, gain)
```

```
grid on
```

```
xlabel('Number of Antenna Elements')
```

```
ylabel('Antenna Gain (dB)')
```

```
title('Antenna Gain for Different Antenna Array Sizes')
```

OUTPUT

